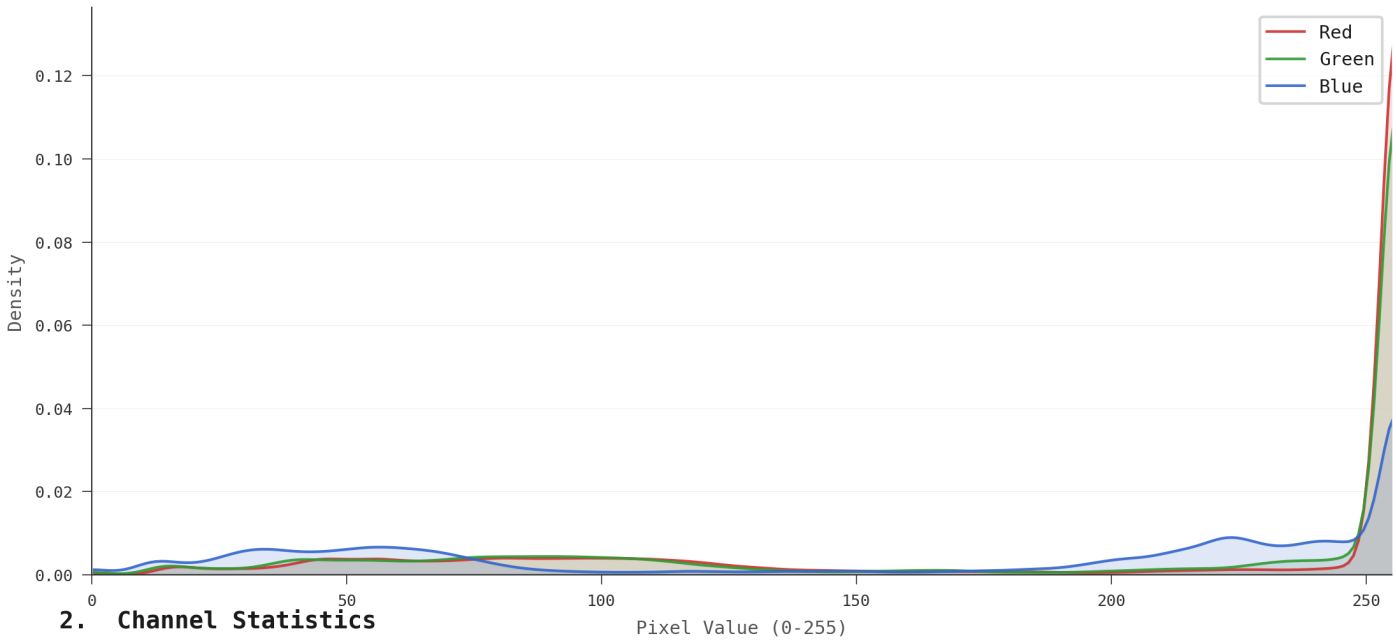
Reference Data Sheet: KODIM20

Subject: P-51 Six-Shooter Aircraft

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	180.5355	86.6838	7514.0811	-1.4473	-0.5175	0	255
Green	176.2617	87.3215	7625.0418	-1.4833	-0.4703	0	255
Blue	154.6573	90.9038	8263.5089	-1.6533	-0.3103	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9955	0.9584
G	0.9955	1.0000	0.9764
B	0.9584	0.9764	1.0000

Pairwise Pearson Correlation Coefficients

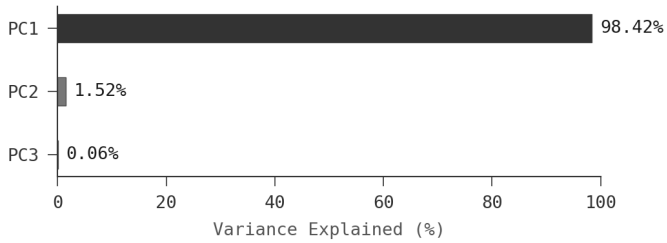
r(R-G)	=	**0.995513**
r(R-B)	=	**0.958368**
r(G-B)	=	**0.976444**
Avg |r|	=	**0.976775**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{7514.1002} & 7535.4115 & 7551.8563 \\
7535.4115 & \mathbf{7625.0611} & 7750.8951 \\
7551.8563 & 7750.8951 & \mathbf{8263.5300}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM20

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5664	-0.5743	**-0.5910**	23033.4748
PC2	-0.5779	-0.2345	0.7817	355.9724
PC3	-0.5875	0.7843	-0.1991	13.2440

6. Derived Metrics

Condition Number

1739.16

(l1/l3) PC1/PC2: 64.71 | PC2/PC3: 26.88

Blue Channel Independence

2.6%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Blue

Loading: -0.5910 | Balanced

Classification: Near-Degenerate (PC1 = 98.42%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.990422	0.986693	0.988557
Green	0.990321	0.986742	0.988531
Blue	0.991471	0.988794	0.990133

8. Average Local Variance (3x3)

Channel	Value
Red	141.6667
Green	142.8929
Blue	129.5373

9. Redundancy Profile

Classification: Near-Degenerate | Condition Number: 1739.16 | PC1: 98.42%

Correlation: R–G pair dominates at r = 0.995513 | G–B: 0.976444 | R–B: 0.958368

Blue Channel: 2.6% independent | 97.4% captured by PC1

Spatial: Avg autocorrelation: 0.989074 | Avg local variance: 138.0323

Eigenvector: Balanced | R (-0.5664), G (-0.5743), B (-0.5910) near-equal loading

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."